🔄 State Transition Testing

Validating the Workflow

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🚀 TL;DR

Systems don't stay the same; they change based on actions.

State Transition Testing checks if the system moves to the **correct** next state (e.g., "Pending" ➡️ "Paid") and blocks **invalid** moves (e.g., "Shipped" ➡️ "Cancelled").

🧐 The Core Concept

Some features have a "Life Cycle." They start at one stage and move to another.

We need to test two things:

1. **Valid Transitions:** Can I go from A to B? (Happy Path).
2. **Invalid Transitions:** Does the system stop me from going from B back to A? (Negative Path).

🚦 The Traffic Light Analogy

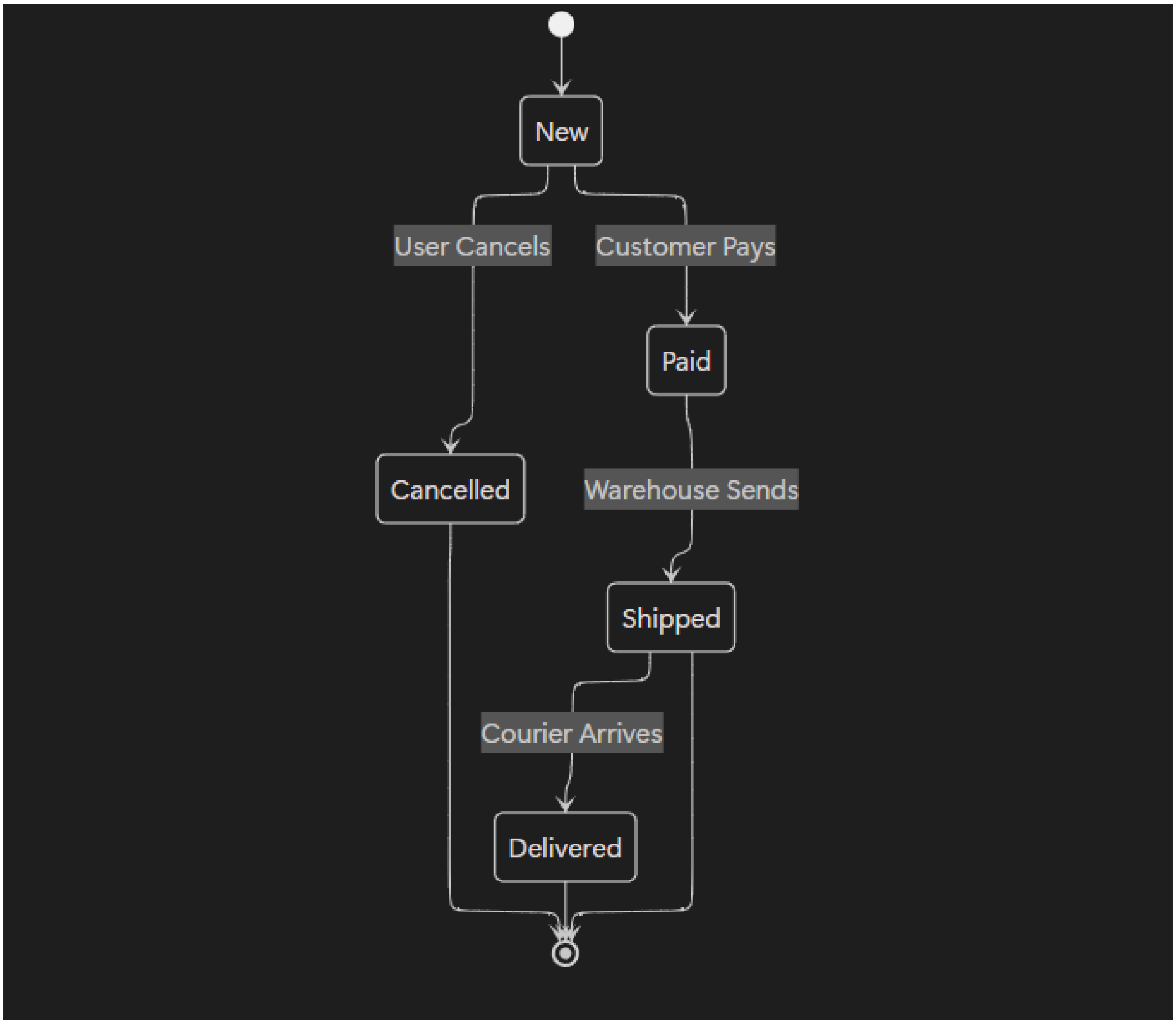
Think of a standard traffic light.

- **Valid:** Green ➡️Orange ➡️ Red.
- **Invalid:** Green ➡️ Red (Dangerous!).

If you are testing the traffic light software, you must prove that it is **impossible** for the light to switch instantly from Green to Red without showing Orange first.

📊 The State Diagram

(This diagram shows the life of an Online Order).



💡 Real World Example: E-Commerce Order

The States:

- **New:** Order created.
- **Paid:** Payment confirmed.
- **Shipped:** Package left warehouse.
- **Cancelled:** Order stopped.

Test Case 1: The Valid Path (Pass)

- *Action:* User pays for a "New" order.
- *Expected Result:* Order Status changes to "Paid".

Test Case 2: The Invalid Path (The Bug Hunter)

- *Scenario:* The user tries to cancel an order *after* it has shipped.
- *Action:* Click "Cancel" on a "Shipped" order.
- *Expected Result:* **Error Message.** (You cannot cancel a shipped item).
- *Bug:* If the status changes to "Cancelled," the company loses money (product is gone, money is refunded).

🧠 Why This Matters

Developers often focus on the "Happy Path" (New ➡️ Paid ➡️ Shipped).

As a Tester, my job is to find the **illegal shortcuts**.

"Can I approve a loan that has already been rejected?"

"Can I log out if I am not logged in?"

State Transition testing catches these logic holes.